

Apartify Africa

KYC & AML Compliance System

Making apartment leasing safe and compliant across Nigeria

Why KYC Matters

- **Regulatory requirement** — CBN (Central Bank of Nigeria) guidelines mandate identity verification for financial services
- **Prevents fraud** — stops money laundering, identity theft, and scam accounts
- **Builds trust** — verified users feel safer transacting on the platform
- **Unlocks features** — wallet withdrawals are blocked until KYC is completed

Key Stat

72%

of fraudulent transactions in Nigeria originate from unverified accounts. KYC is the first line of defense.

Verification Tiers

Four-tier system — withdrawals are gated until **Basic** tier or higher.



Unverified

Default

Browse listings, make bookings. **No withdrawals.**



Pending

Under Review

Documents submitted, awaiting admin review.



Basic

Approved

Can bind bank account and **withdraw funds.**



Verified

Full Access

Complete verification. **All features unlocked.**

User KYC Flow



Accepted Document Types

- NIN (National ID)
- Voter's Card
- International Passport
- Driver's License

Bank Account Binding

How it works

1. User provides: **Bank name, Account number, Account holder name**
2. Backend verifies the account against **Flutterwave** (Nigerian bank API)
3. Account is bound to the user's profile
4. **Only bound accounts** can receive withdrawals

Why? Prevents funds from being sent to accounts the user doesn't own. Critical for AML compliance.

Requirements

Requirement	Detail
KYC Tier	Basic or higher
Verification Provider	Flutterwave
Bank Code	Required (e.g., 044 — Access Bank)
Account Number	10-digit Nigerian account
Account Name	Must match Flutterwave response

Admin Compliance Dashboard

Review KYC

- View pending submissions queue
- See document type and masked number
- Approve → set tier (Basic / Verified)
- Reject → provide reason

Wallet Management

- Freeze/unfreeze wallets
- Suspend suspicious accounts
- Block deposits and withdrawals
- Reason logged with admin identity

AML Flags

- Auto-generated compliance alerts
- Severity grading (Low → Critical)
- Resolve with optional notes
- Full audit trail

Admin Endpoints

```
// KYC Review
GET    /api/admin/compliance/kyc/pending    // Submissions awaiting review
GET    /api/admin/compliance/kyc           // All submissions
PUT    /api/admin/compliance/kyc/{userId}/approve // Approve KYC
PUT    /api/admin/compliance/kyc/{userId}/reject  // Reject KYC

// Wallet & Flags
PATCH /api/admin/compliance/wallets/{userId}/status // Freeze/Unfreeze
```

GET /api/admin/compliance/flags // AML alerts

PATCH /api/admin/compliance/flags/{id}/resolve // Resolve flag

AML Compliance Flags

The system automatically raises flags for suspicious activity, graded by severity.

Flag Type	Trigger	Severity
Withdrawal to Unbound Account	User tries to withdraw to an unverified account	HIGH
Bank Binding Mismatch	Bank details don't match Flutterwave verification	HIGH
Deposit Limit Exceeded	Single deposit above threshold	MEDIUM
Withdrawal Limit Exceeded	Single withdrawal above threshold	HIGH
Withdrawal Velocity Exceeded	Too many withdrawals in short period	MEDIUM
Recent Deposit + Withdrawal	Funds withdrawn shortly after deposit (layering)	HIGH
Large Transaction	Unusually large amount detected	MEDIUM
KYC Rejected / Approved	Audit trail for verification decisions	LOW

Security & Data Protection

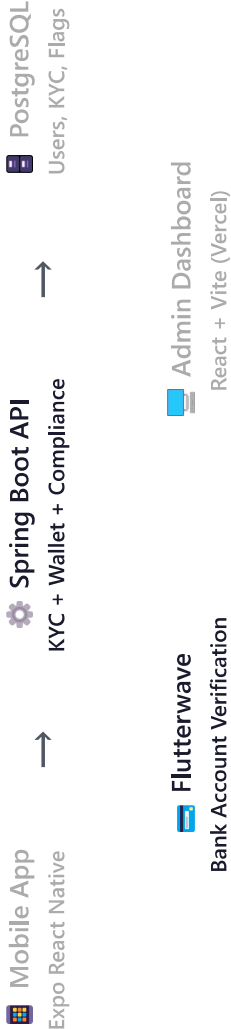
Data Protection

- 🔒 Document numbers are **masked** in the database
- 🔒 Only the **last 4 digits** stored in plain text
- 🔒 Full numbers are **never logged** or returned in API responses
- 🔒 Bank verification done **server-side** via Flutterwave — no account numbers in frontend

Audit & Compliance

- 📋 All admin actions are **audit-logged**
- 📋 Each log includes: admin email, timestamp, action, and affected entity
- 📋 KYC decisions (approve/reject) create compliance flag records
- 📋 Wallet status changes generate audit entries with reason

Technical Architecture



Key Files

```
// Backend (Spring Boot)
apps/api/.../service/KycService.java           // Core KYC logic
apps/api/.../controller/KycController.java      // User-facing endpoints
apps/api/.../controller/AdminComplianceController.java // Admin review & AML
apps/api/.../entity/ComplianceFlag.java        // Compliance flag entity

// Mobile (Expo)
apps/mobile/src/services/kycService.js         // KYC client service
apps/mobile/src/screens/WalletScreen.js        // KYC banner + withdrawal gate
```


Implementation Summary

- ✓ Multi-tier KYC (Unverified → Verified) ✓ AML compliance flags (auto-detection)
- ✓ Document submission & admin review ✓ Wallet freeze/unfreeze by admin
- ✓ Bank account binding via Flutterwave ✓ Audit logging for all admin actions
- ✓ Withdrawal gating by KYC tier ✓ Document number masking

Next Steps

Document image upload · Automated BVN verification · Real-time admin notifications · Batch review